If heat is evolved by the system : -q

If work is done by the system : -w

If work is done on the system : +w

Laws of Thermodynamics:

Zeroth law : When two systems are separately in equilibrium with a third system, they are in equilibrium with each other.

First law: "Energy can neither be created nor destroyed, but may be converted from one form to another".

$\Delta U = q + w$.

Enthalpy is a thermodynamic property of a system. Enthalpy H is defined as the sum of the internal energy and pressure volume work.

$H=U+PV$. Enthalpy change, $\Delta H = \Delta U + \Delta n_g RT$.

Hess's law: It states that "the enthalpy change of a reaction either at constant volume or constant pressure is the same whether it takes place in a single or multiple steps". Hess's law can be applied to calculate enthalpies of reactions that are difficult to measure.

Heat capacities: [C_p and C_v]

Heat capacity is defined as the amount of energy required to increase the temperature of one unit quantity of material by one degree, under specific conditions. It can be measured under two different conditions, namely,

(a) constant pressure $C_p = (dH/dT)_p$

(b) constant volume $C_v = (dU/dT)_v$

Second law of thermodynamics:

The second law of thermodynamics helps us to predict whether the reaction is feasible or not and also tells the direction of the flow of heat.

To predict spontaneity of a process, a new thermodynamic quantity namely entropy (S) was introduced. Entropy is a measure of the randomness or disorderliness of the system.

Entropy statement: "whenever a spontaneous process takes place, it is accompanied by an increase in the total entropy of the universe".

Kelvin-Planck statement: It is impossible to take heat from a hotter reservoir and convert it completely into work by a cyclic process without transferring a part of that heat to a colder reservoir.

Clausius statement: This statement recognises that heat flows spontaneously from hot objects to cold objects and to get it flow in the opposite direction, we have to spend some work.

Gibbs Free Energy (G):

G is expressed as $G=H-TS$, free energy change of a process is given by the relation $\Delta G= \Delta H-T\Delta S$.

Standard free energy change and equilibrium constants are related by the equation $\Delta G^0 = -RT\ln K_{eq}$

Third law:

The entropy of a perfectly crystalline material at absolute zero is zero.

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